

Working paper draft — AI-for-finance / systems framing (sibling of the JF/RFS draft)

A Market World-Model Runtime for AI Consumers: Compiling Asynchronous Evidence into Analyzable, Abstention-Aware, and Tradeable State

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Canonical TeX: pdf/sci/main_ai_wm.tex. JF/RFS sibling: pdf/sci/main_jf_rfs.tex. Thesis: a financial-market world-model runtime compiles the raw market into an LLM-analyzable, abstention-aware, tradeable state; trading numbers prove the world has content—they do not proclaim a strategy Holy Grail.

Abstract

Large language models and autonomous agents increasingly *consume* market evidence, but raw feeds are asynchronous, intermittently missing, and quality-heterogeneous. What an AI needs is not another feature dump: it needs a *world-model runtime* that compiles raw evidence into a state that can be analyzed, refused when thin, and acted on when ready. We develop RCA-WM: epistemic observations $O_{j,t}=(x,\tau,q,g,r)$, a compilation operator Π_t , reconstruction bounds, WMI/ACWMI with world-conditional abstention, and a LOBO content/gating decomposition. The system contribution is a point-in-time multi-band laboratory that exposes quality-tagged bundles, availability shocks O_t , and a Compiled-versus-Raw consumer protocol.

Empirically, on a PIT panel (399 days; durable bands: exchange, macro, alternative), vintaged macro and stablecoin-flow *content* changes a transparent action rule relative to same-input momentum (ΔCE 0.334, $p = 0.034$; relative gap survives 10–25 bps costs). LOBO attributes losses mainly to content; a 2017–2026 audit finds no hidden return-rule alpha when band archives are absent. **Thesis:** we build a financial-market world-model runtime that compiles the raw market into an LLM-analyzable, abstention-aware, tradeable state; trading numbers prove that this world has content—they do not proclaim a strategy Holy Grail.

Keywords: world models; AI for finance; information-set compilation; selective prediction; point-in-time; cryptocurrency; LLM consumers. **JEL:** G12, G14, C58, C55

1. Introduction

AI systems that analyze or trade markets do not observe a clean panel. They face venue fragmentation, perpetual leverage, options walls, unlocks, on-chain flows, news, and macro vintages on incompatible clocks. Asset-pricing and ML-pricing literatures typically treat the information set I_t as given and study how it is *used*. For AI consumers the prior engineering problem is different: how I_t is *compiled* into a usable world state, when that world is too thin to act, and how the compiled object is served to an LLM or policy.

Thesis. We build a financial-market *world-model runtime*: it compiles raw market evidence into a state that an LLM (or any AI consumer) can analyze, abstain on, and—when warranted—trade. Trading statistics are instruments that show the compiled world has nonempty content; they are not a claim of unconditional strategy alpha.

Three contributions, ordered for an AI-for-finance / systems audience. **World-model theory:** epistemic observations, Π_t , reconstruction bounds, WMI/ACWMI abstention, and LOBO content/gating; the SDF / compilation-wedge interface is a consistency remark for agents constrained to $\blacksquare^{AI}_t$, not the headline estimand. **System (runtime):** collectors, BandPIT, readiness/WMI/ACWMI, O_t , paper snapshots, AI bundles, and a Compiled-versus-Raw consumer protocol. **Content validation:** mechanism – momentum on the PIT panel shows compiled band content is economically nonempty (ΔCE 0.334, $p = 0.034$; LOBO content-dominant)—not a trading Holy Grail and not live LLM alpha.

2. Related literature

Information sets in asset pricing. Classic and modern frameworks treat I_t as given (Cochrane 2005; Fama–French 1993; Gu–Kelly–Xiu 2020; Kelly–Pruitt–Su 2019; Nagel 2021; Harvey–Liu–Zhu 2016). Machine-learning asset pricing expands the feature span inside I_t but typically assumes synchronized panels. We study the prior step: compilation.

Cryptocurrency market structure. Fragmentation and distinctive risk factors (Makarov–Schoar 2020; Liu–Tsyvinski–Wu 2022) motivate multi-band worlds rather than price-only predictors.

Selective prediction and measurement timing. When the world is thin or dishonest, abstention can be optimal. Macro vintages and crypto freshness/missingness make measurement timing first-class.

3. Theory

Reading guide (AI-for-finance). The core object is runtime semantics: epistemic observations $\rightarrow$ operator Π_t $\rightarrow$ reconstruction bounds $\rightarrow$ WMI/ACWMI and abstention $\rightarrow$ LOBO content/gating $\rightarrow$ AI bundle schema. The SDF / compilation-wedge paragraph is a pricing-consistency interface for agents that condition only on $\blacksquare_t^{\text{AI}}$. Section 7 validates that compiled content is nonempty; it does not redefine the contribution as a pure asset-pricing horse-race.

3.1 World-model quality: breadth, stability, honesty

Let K be the number of critical evidence bands and $a_{k,t} \in \{0,1\}$ an AI-usable indicator. Breadth, stability, and honesty are:

$$B_t = (1/K) \sum_k a_{k,t}, \quad U_t = \exp(-\sum_j \omega_j d_{j,t}), \quad H_t = 1 - (1/J) \sum_j m_{j,t}.$$

The production baseline World Model Index is the product form

$$\text{WMI}_t = B_t \times U_t \times H_t.$$

Hierarchical breadth and continuous honesty used in the PIT panel:

$$B_t^{\text{hier}} = 0.25 B_t^{\text{dom}} + 0.35 B_t^{\text{band}} + 0.40 B_t^{\text{asset}},$$

$$H_t^{\text{cont}} = \exp(-2c_t) \max(0, 1 - 0.5(1 - e_t)),$$

where e_t is the ready-share and c_t a contamination share.

3.2 Epistemic observations

An AI-consumable market observation is not a bare scalar:

$$O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t}),$$

with value, latest-available time, quality, main-view gate, and semantic role. Numerically identical x with different (τ, q, g, r) are different world-model objects.

3.3 Asynchronous state space and lag error

Latent states evolve as $S_{t+1} = F(S_t, \eta_{t+1})$. Source j observes a lagged map $X_{j,t}^{\text{obs}} = h_j(S_{t-\blacksquare}) + v_{j,t}$. If h_j is Lipschitz, reconstruction error admits a bound with delay, noise, and missingness terms:

$$\blacksquare S_t - S_t \blacksquare \leq C_1 \sum \omega_j \blacksquare_{j,t} + C_2 \sum \omega_j \blacksquare_{j,t} v_{j,t} \blacksquare + C_3 \sum \omega_j (1 - z_{j,t}).$$

Latest-snapshot, freshness/TTL, and readiness gates target these three terms.

3.4 Filters and the compilation operator

$$F_t^{\text{raw}} = \sigma(\{X_{j,\tau}^{\text{obs}}\}), \quad F_t^{\text{AI}} = \sigma(W_t^{\text{AI}}, D_t), \quad \Pi_t = B_t \blacksquare M_t \blacksquare A_t, \quad W_t^{\text{AI}} = \Pi_t(F_t^{\text{raw}}).$$

Proposition 1 (compilation $\neq$ feature expansion). Enlarging F^{raw} without a well-defined Π need not enlarge decision-relevant F^{AI} : ungated, stale, or role-incoherent evidence can expand raw span while shrinking usable world quality via H_t and U_t .

Proof sketch. Add a source j^* with large lag or missingness flag. If M_t fails to exclude it, H falls by $\Delta > 0$ and freshness penalties weakly lower U . Breadth rises by at most $1/K$, but multiplicative $\text{WMI} = \text{BUH}$ can fall when UH losses dominate. Since $W^{\text{AI}} = \Pi(F^{\text{raw}})$ inherits gated objects, raw σ -field expansion need not enlarge payoff-relevant AI-visible information. ■

Proposition 2 (SDF interface: compilation constrains pricing). An agent whose usable conditioning information is F^{AI} can only implement pricing statements $E[mR | F^{\text{AI}}] = 1$. Evaluating the same agent against the finer F^{raw} produces a *compilation wedge* $w_t = E[mR | F^{\text{raw}}] - E[mR | F^{\text{AI}}]$, which is zero for all payoffs iff compilation loses no pricing-relevant information. Alpha measured against the raw filtration confounds skill with compilation loss.

Proof sketch. By iterated expectations $E[w_t | F^{\text{AI}}] = 0$, but $w_t \neq 0$ pointwise unless the raw-conditional moment is F^{AI} -measurable. If a band with $\text{MIG} > 0$ is discarded, a payoff exists whose raw-conditional expectation varies with that band while the compiled one does not. ■

Proposition 3 (lag reconstruction bound). Under Lipschitz observation maps and bounded increments $\blacksquare S_u - S_{u-1} \blacksquare \leq \delta \blacksquare$, reconstruction error admits the delay / noise / missingness bound above with $C' \blacksquare = C \blacksquare \delta \blacksquare$ independent of the AI model class.

Proof sketch. Lipschitz gives $\blacksquare h_j(S_t) - h_j(S_{t-\blacksquare}) \blacksquare \leq L_j \delta \blacksquare$. Decompose $S \blacksquare - S$ into lagged-map bias, v , and missingness; triangle inequality + weights ω_j yield the bound. TTL / readiness operators shrink exactly those three terms. ■

3.5 ECP, MIG, and causal DAG

$$\text{ECP}_t = 1\{\text{conf}_t > c \blacksquare\} 1\{\text{WMI}_t < w\}.$$

Thick versus thin information sets differ by conditional mutual information; band k 's marginal information gain on task m is $\text{MIG}_{k,t}^{(m)} = I(R_t^{(m)}; E_{k,t} | I_t^{(-k)})$.

Identification DAG: $O_t \rightarrow W_t \rightarrow A_t$, with market complexity M_t and latent configuration C_t as confounders. Availability shocks O_t are the preferred quasi-exogenous lever.

3.6 Bayesian abstention

With action set $A = \{\text{bullish}, \text{bearish}, \text{neutral}, \text{abstain}\}$,

$$a_t^* = \arg \min_a E[\blacksquare(a, R_t) | W_t].$$

Abstain when every non-abstain action has expected loss above a world-dependent cost $c_{\text{abs}}(W_t)$.

Proposition 4 (world-conditional abstention). If $\blacksquare(\text{abstain}) \equiv c_{\text{abs}}(W)$ and every non-abstain action has $E[\blacksquare | W] > c_{\text{abs}}(W)$, then $a^* = \text{abstain}$. Under a single-crossing assumption (now formalized as Assumption 1), with non-abstain loss and c_{abs} weakly decreasing in WMI , the abstention region is a lower set in WMI (implemented by IS-frozen ACWMI thresholds).

Proof sketch. First claim is the Bayes argmin. Second: let $L \blacksquare(W) = \min_{a \neq \text{abs}} E[\blacksquare | W]$; single crossing makes $\{W: L \blacksquare > c_{\text{abs}}\}$ a lower contour set. ■

3.7 ACWMI and explanation metrics

$$\text{ACWMI}_t = \exp(\sum_i \gamma_i(r_t) \log x_{i,t} / \sum_i \gamma_i), \text{ with } x = (B^{\text{hier}}, U, H^{\text{cont}}, S, C).$$

$\text{EAR}_t = (\# \text{ evidence-bound claims})/(\# \text{ claims})$, $\text{UCR}_t = 1 - \text{EAR}_t$, $\text{EV}_t = d(\Phi_t, \Phi_{t-1}) / (1 + d(W_t, W_{t-1}))$. Thin and thick worlds also differ in explanation sets Φ_t .

Proposition 5 (ACWMI factor monotonicity). With $\gamma \neq 0$ and $x \in (0,1]$, $\partial \text{ACWMI} / \partial x_k > 0$ and $\log \text{ACWMI}$ is concave in $\log x$. A proportional honesty deterioration cannot be offset one-for-one by raising breadth with equal weight.

Proof sketch. $\log \text{ACWMI} = \sum w_i \log x_i$ with $w_i = \gamma_i / \sum \gamma_i$; $\partial \text{ACWMI} / \partial x_k = \text{ACWMI} \cdot w_k / x_k > 0$. Under contamination, $\Delta \log H$ is large while $\Delta \log B = O(1/K)$, so net $\sum w \Delta \log x < 0$. ■

Proposition 6 (LOBO as economic MIG, with channel decomposition). Let $V(I)$ be OOS CE of a fixed rule on information set I and $\text{MIG}_k = V(I) - V(I \setminus E_k)$. Under Blackwell monotonicity and non-redundancy, $\text{MIG}_k > 0$. Band k enters through a *content* channel (its signal changes the action) and a *gating* channel (its availability changes B/U/H and abstention); the MIG telescopes exactly into the two channel effects. In the PIT laboratory, forcing the band tilt to zero implements content deletion; setting band status to missing implements gating deletion; both together implement $I \setminus E_k$.

Proof sketch. Blackwell $\Rightarrow V(I) \geq V(I \setminus E_k)$; strict when E_k has unique predictive content. The decomposition is a telescoping identity over the three experiment arms (content-only, gating-only, both). ■

Proposition chain (summary). Prop.1 separates raw span from usable world quality; Prop.2 constrains any pricing kernel to the compiled filtration (alpha vs compilation wedge); Prop.3 grounds freshness/TTL as reconstruction controls; Prop.4 justifies world-conditional abstention; Prop.5 makes ACWMI a strict quality index; Prop.6 maps LOBO CE drops to economic MIG and separates content from gating. Empirics below instantiate this chain—they do not replace it.

4. The EvoQuant laboratory as measurement instrument

EvoQuant supplies multi-band collectors, readiness scoring, BandPITService / time_slice PIT reconstruction, availability-shock queries over collection_runs, and configurable WMI/ACWMI thresholds (WORLD_MODEL_INDEX_MODE, WMI_ABSTAIN_THRESHOLD, ACWMI_ABSTAIN_THRESHOLD). Epistemological order: theory first, laboratory second.

5. Data and real multi-band PIT archive

5.1 Populated archive

Exchange (OKX): ~3200 klines/merged bars, daily 2025-07-01 → 2026-08-04; ~0 funding rows. Macro: 85617 vintaged points. Alternative: 88803 points. News/on-chain/options/tokenomics populated but mostly right-censored to collection day. Analytics snapshots remain sparse historically; multi-band PIT uses raw history tables via `build_pit_archive.py` (production path prefers `BandPITService`).

5.2 PIT panel

Timing protocol: decision at previous close. For payoff day t , band statuses are evaluated at $(t-1)$ 23:59; content tilts use vintaged `available_at` with a one-day lag. Panel: 2025-07-01 → 2026-08-03, 399 days $\times$ 10 assets = 3990 rows. Durable bands: exchange, macro, alternative. Yahoo daily returns provide payoffs; engines use only pre- t returns.

Empirical ready rates (approx.): `exchange=0.803`, `news=0.000`, `event_calendar=0.000`, `onchain=0.000`, `tokenomics=0.000`, `options=0.000`, `alternative=1.000`, `macro=1.000`.

6. Empirical design

Chronological IS/OOS split at 2026-01-16 (200/200 days). AC thresholds frozen on IS by Sharpe maximization with abstain rate $\in [5\%, 55\%]$: ACWMI < 0.25 or $C < 0.25$ (calibration selects minimal gating). Production WMI threshold 0.2 is never tuned on OOS. Policies: always-long; momentum; mechanism with band content; simple outage; cascade; WMI; ACWMI (IS-frozen). Economic value: annualized return/vol, Sharpe, CRR CE ($\gamma=2$), max DD. **Inference:** circular block bootstrap on OOS daily PnL ($n_{\text{boot}}=999$, $\text{block}=5$) with block-length sensitivity $\{5,10,21\}$ and a White (2000) reality check; costs 10/25/50 bps on turnover plus perp-funding adjustment. Identification: thin vs thick (content+gating); LOBO with channel decomposition; scarce-world event study; 2017–2026 long-span audit.

6.1 Mechanism signals: transparent rule with band content

Directional actions are a deterministic rule on named production calculators, and band *content* enters directly: `macro_tilt` (vintaged VIX/DXY 5d changes: both down = risk-on +1, both up = risk-off -1) and `alt_tilt` (stablecoin 7d net-supply sign, latest obs $< t$). Tilts are forced to 0 when the band is not PIT-ready, so LOBO deletes content, not only gating. Rule: $R1 \text{ crisis} \wedge \text{cascade_p} \geq 0.60 \rightarrow \text{short}$ (evidence conjunction); $R2 \text{ trend} \wedge \text{mom5} > 0 \wedge \text{cascade_p} < 0.45 \wedge \text{macro_tilt} \geq 0 \rightarrow \text{long}$; $R2b \text{ range} \wedge \text{both tilts} > 0 \wedge \text{mom5} \geq 0 \rightarrow \text{long}$; $R3 \text{ sign}(\text{mom5})$ with a double-risk-off long veto ($\rightarrow \text{flat}$), ties broken by $\text{sign}(\text{tilt sum})$. C = pairwise sign agreement among $\{\text{mom5}, \text{flow}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}, \text{macro_tilt}, \text{alt_tilt}\}$.

Calibration audit and engine-input repair (disclosed). A long-span audit exposed two harness flaws: full-history drawdowns made crisis an absorbing state, and the cascade input map saturated at 0.86 (a constant short that won a bear OOS by construction). Repaired with 60-day windows, volatility-standardized tail intensity, and real RSI/ADX proxies; `cascade_p` is now monotone in realized next-day left tails ($P(r < -5\%)$ rising 0.039 $\rightarrow$ 0.110 across deciles). The repair was driven by the audit, not by OOS performance search.

component	formula	inputs	role
<code>cascade_p</code>	<code>LiquidationCascadeCalculator.compute_cascade_probability(cluster, capacity, distance)</code>	left-tail cluster intensity from pre-t returns; capacity=6e7; distance by vol regime	crisis / short trigger in R1
<code>systemic</code>	<code>ContagionRiskCalculator.compute_systemic_risk_score([covar_95, corr, tail_beta])</code>	asset vs synthetic peer path from pre-t returns	sign in consistency C
S (signal integrity)	<code>clip(hl_factor*(1-0.7*crowding_n)*(0.35+0.65*surprise_n))</code>	AlphaDecay half-life on <code>cumsum(returns)</code> , crowding, surprise of last return	ACWMI factor
C (consistency)	pairwise sign-agreement among $\{\text{mom5}, \text{flow_sign}, -1_{\{\text{casc} > 0.55\}}, -1_{\{\text{sys} > 55\}}\}$	<code>mom5</code> , VPIN/flow class, <code>cascade_p</code> , <code>systemic</code>	ACWMI factor + AC abstention gate
<code>detected_regime</code>	<code>RegimeClassifier.classify_price_regime(RegimeFeatures)</code> with crisis override	returns, rolling vol, RSI/ADX proxies, <code>cascade_p</code> , <code>vol_regime</code>	R1/R2 branching
<code>mom5</code>	<code>sign(mean(r_{t-5:t}))</code>	last 5 pre-t daily returns	R2/R3 directional fallback
<code>macro_tilt</code>	+1 if <code>VIX_5d_chg</code> < 0 and <code>DXY_5d_chg</code> < 0; -1 if both > 0; else 0; forced 0 when macro band not ready	vintaged <code>macro_timeseries</code> (available_at $\leq t$): VIX, DXY	macro-band content: veto in R2, band-long in R2b, tie-break in R3, sign in C
<code>alt_tilt</code>	<code>sign(stablecoin_net_supply_change_7d)</code> at latest obs $< t$; forced 0 when alternative band not ready	<code>alternative_timeseries</code> stablecoin 7d net supply (pre-t)	alternative-band content: band-long in R2b, tie-break in R3, sign in C
<code>signal</code>	$R1 \text{ (crisis AND casc} \geq 0.60) \rightarrow -1$; $R2 \rightarrow +1$ (macro veto); $R2b \rightarrow +1$ (band-driven); $R3 \text{ sign}(\text{mom5})$ with double-risk-off long veto; ties $\text{sign}(\text{macro_tilt} + \text{alt_tilt})$	<code>detected_regime</code> , <code>cascade_p</code> , <code>mom5</code> , <code>macro_tilt</code> , <code>alt_tilt</code>	position when not abstaining; R1 requires evidence conjunction

Table. Mechanism component definitions (audit).

7. Main results

7.1 Band content beats momentum out of sample

Always-long loses badly in this bear window (CE -1.157); momentum is nearly flat (Sharpe 0.101 , CE -0.202); the transparent rule with band content earns Sharpe 0.767 and CE $+0.132$, standing aside on 7.5% of asset-days via content-driven vetoes. IS-frozen ACWMI is secondary (Sharpe 0.654 , CE $+0.076$). **Headline: mechanism – momentum $\Delta\text{CE} = 0.334$, $p = 0.034$, 95% CI $[0.03, 0.68]$ excludes zero** (first row of the bootstrap table). The two policies share all return-based inputs and differ only in vintaged macro/alternative content.

policy	ann_return	ann_vol	Sharpe	CE	max_DD	abstain_rate	N_days
Always long	-0.8164	0.5836	-1.399	-1.1568	-0.4661	0.0	200
Momentum always	0.0506	0.5009	0.101	-0.2019	-0.4997	0.0	200
Thick ungated	0.3846	0.5012	0.767	0.132	-0.4378	0.075	200
Simple outage rule	-0.0712	0.4432	-0.161	-0.2699	-0.4378	0.438	200
Simple cascade rule	0.3846	0.5012	0.767	0.132	-0.4378	0.075	200
WMI threshold (0.2)	0.0	0.0	0.0	0.0	0.0	1.0	200
ACWMI (IS-frozen)	0.3246	0.4967	0.654	0.0764	-0.4378	0.127	200

Table 1. OOS economic value on real PIT panel (200 days).

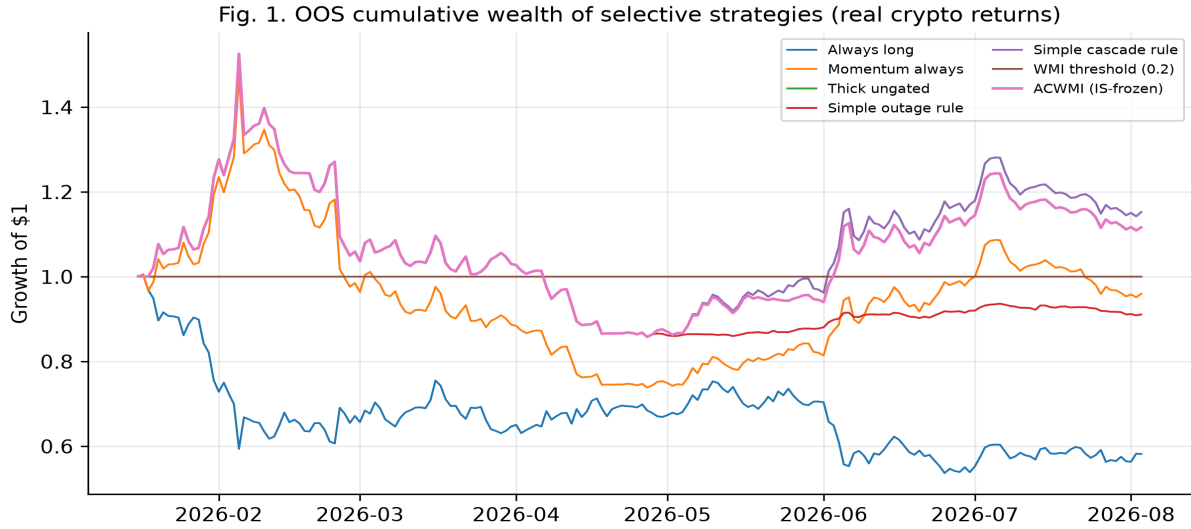


Figure 1. OOS cumulative wealth.

Fig. 2. Economic value of abstention policies (OOS)

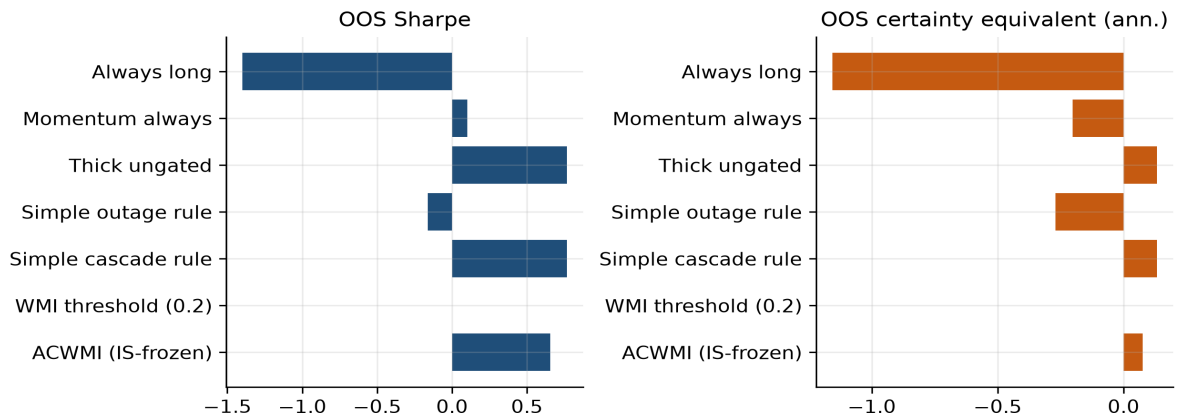


Figure 2. OOS Sharpe and CE by policy.

7.2 Thick vs thin worlds

The thin exchange-only observer (content AND gating deleted) abstains 61% and earns CE -0.404 ; the thick world earns $+0.132$. $\Delta\text{Sharpe} = 1.78$ ($p = 0.036$, CI excludes zero); $\Delta\text{CE} = 0.54$ ($p = 0.24$, not yet significant at 200 days).

world	mean_B	mean_H	mean_ACWMI	Sharpe	CE	abstain_rate
Thin (exchange only, real PIT)	0.123	0.538	0.165	-1.011	-0.4037	0.614
Thick real PIT (ex+macro+alt...)	0.277	0.662	0.339	0.767	0.132	0.075
Thick gated AC (real PIT)	0.277	0.662	0.339	0.654	0.0764	0.127

Table 2. Thin vs thick on the real PIT archive (OOS).

Fig. 6. Thin vs thick-ungated vs thick-gated worlds (OOS)

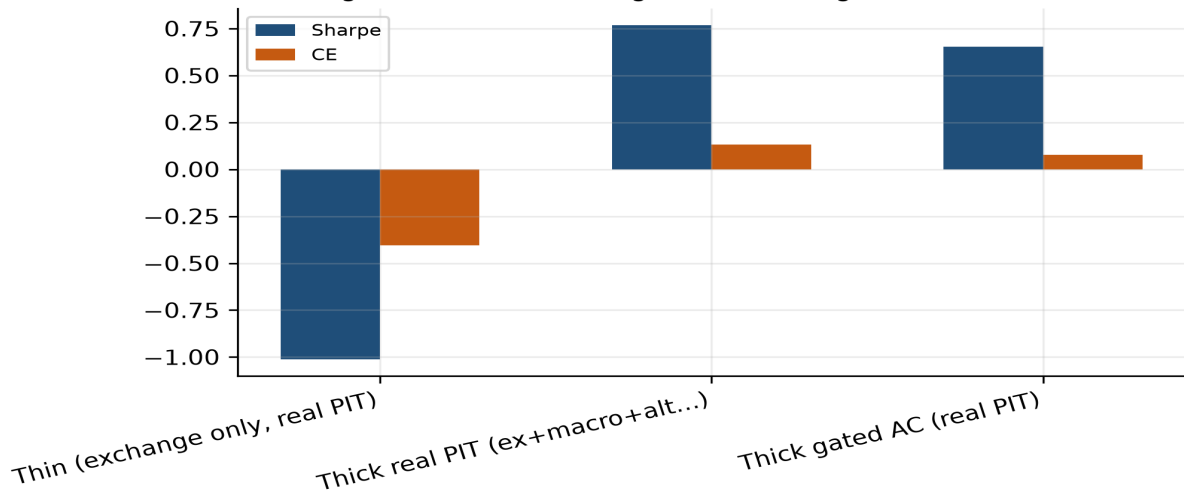


Figure 3. Thin vs thick worlds (real PIT).

7.3 Leave-one-band-out: significant, and driven by content

Under the headline ungated mechanism, deleting macro or alternative costs $\Delta\text{CE} -0.334$ ($p = 0.034$) — exactly the mechanism — momentum gap — because content deletion collapses the rule onto momentum. Exchange status deletion costs 0 under the ungated rule. The telescoping identity holds exactly: content share of

total loss is 1.0 and the gating residual after content deletion is 0. Band information changes *what* the rule trades, not merely *when* it abstains.

band_dropped	Sharpe	CE	abstain_rate	dCE	p_dCE	policy
(none)	0.767	0.132	0.075	0.0		thick_ungated
exchange	0.767	0.132	0.075	0.0	1.0	thick_ungated
macro	0.101	-0.2019	0.0	-0.3339	0.034	thick_ungated
alternative	0.101	-0.2019	0.0	-0.3339	0.034	thick_ungated
macro+alternative	0.101	-0.2019	0.0	-0.3339	0.034	thick_ungated

Table 3. Leave-one-band-out on durable PIT bands (total effect).

band	dCE_total	p_total	dCE_content	p_content	dCE_gating_residual	p_gating_residual	dCE_gating_standalone	p_gating_standalone	telescope_on	content_share_of_total
exchange	0.0	1.0			0.0	1.0	0.0	1.0	True	
macro	-0.3339	0.034	-0.3339	0.034	0.0	1.0	0.0	1.0	True	1.0
alternative	-0.3339	0.034	-0.3339	0.034	0.0	1.0	0.0	1.0	True	1.0
macro+alternative	-0.3339	0.034	-0.3339	0.034	0.0	1.0			True	1.0

Table 4. LOBO channel decomposition: content vs gating.

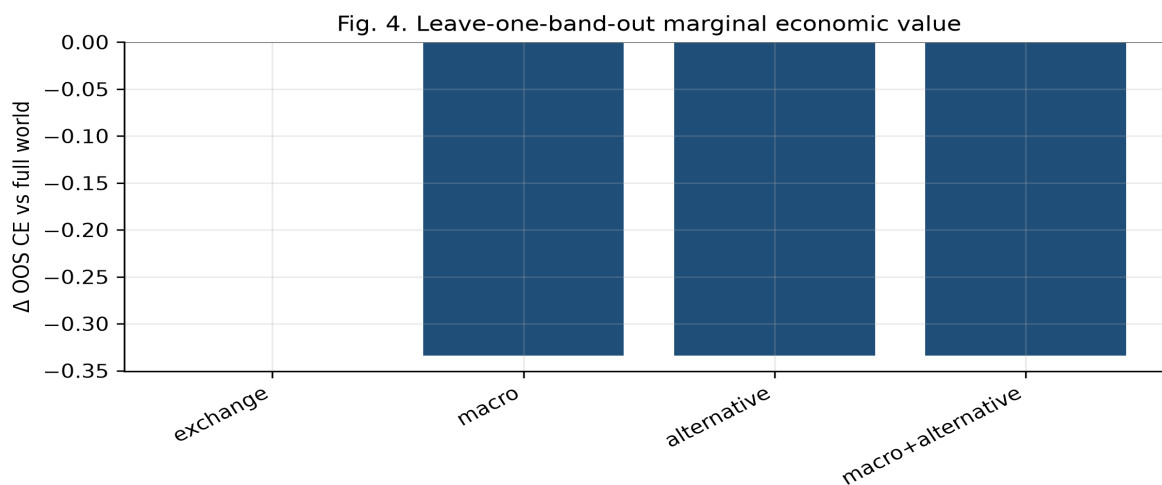


Figure 4. LOBO marginal CE on durable bands.

7.4 Long-span external-validity audit (2017–2026)

The identical rule (tilts zero: no band archive before 2025) on BTC/ETH over 3,471 days: annualized return 0.456, Sharpe 0.664, CE -0.009 , positive in 8 of 10 years including 2018 (+2.13 in a -1.08 always-long year) — and no significant advantage over momentum (Δ CE 0.017, $p = 0.55$). The return-based core has no hidden alpha; the significant OOS gains load on compiled band content, which exists only where the multi-band archive exists.

year	n_days	crisis_share	mech_ann_ret	mech_Sharpe	long_ann_ret	long_Sharpe
2017	334	0.916	1.2976	1.379	3.6928	3.916
2018	365	0.986	2.1252	2.639	-1.0754	-1.2
2019	365	0.752	0.8027	1.242	0.582	0.832
2020	366	0.54	0.6022	0.788	1.944	2.435
2021	365	0.968	0.4554	0.561	1.4869	1.689
2022	365	0.995	-0.9128	-1.316	-0.7852	-1.075
2023	365	0.426	0.1539	0.374	0.8933	2.066
2024	366	0.71	0.1622	0.302	0.7598	1.35

year	n_days	crisis_share	mech_ann_ret	mech_Sharpe	long_ann_ret	long_Sharpe
2025	365	0.668	-0.2523	-0.481	0.0926	0.165
2026	215	0.805	0.0247	0.049	-0.5074	-0.944

Table 5. Long-span audit by year (BTC/ETH; crisis_share is the classifier's label share).

casc_decile	n	mean_cascade_p	mean_ret_next	p_tail2	p_tail5
0	1989	0.2	0.00076	0.16	0.039
1	2357	0.309	0.00309	0.232	0.083
2	293	0.326	-0.00126	0.256	0.072
3	666	0.356	0.00349	0.215	0.066
4	660	0.403	0.00081	0.255	0.082
5	663	0.496	-0.00056	0.267	0.11

Table 6. cascade_p calibration: deciles vs realized next-day left tails.

7.5 Bootstrap contrasts, costs, and inference robustness

Block-bootstrap (999×5 -day) contrasts: the mechanism – momentum content contrast is significant ($p = 0.034$); contrasts against always-long remain wide at 200 days. The relative ΔCE stays in $[0.329, 0.334]$ across CRRA $\gamma \in \{1, 2, 4, 6\}$ and survives 10–25 bps costs; leave-one-asset jackknife keeps $\Delta CE \in [0.302, 0.352]$. Stationary bootstrap confirms $p = 0.022$. White (2000) reality check vs always-long: $p = 0.144$. Compilation-wedge sign-projection raises CE by 0.992 vs thin momentum ($p = 0.020$) even though linear R^2 stays near zero — value is in implementable signed payoffs, not predictive R^2 .

contrast	dSharpe	dCE	p_CE	ci_dCE_05	ci_dCE_95
Mechanism – Momentum	0.6664	0.3339	0.034	0.028	0.6768
Mechanism – Always long	2.1664	1.2887	0.246	-0.9963	3.5152
ACWMI – Always long	2.0525	1.2332	0.264	-1.1011	3.4577
ACWMI – Momentum always	0.5526	0.2783	0.104	-0.057	0.6342
Thick ungated – ACWMI	0.1139	0.0556	0.232	-0.0336	0.1673
ACWMI – WMI threshold (0.2)	0.6535	0.0764	0.906	-1.2364	1.2439

Table 7. OOS block-bootstrap contrasts.

policy	cost_bps	funding	Sharpe	CE
Thick ungated	0.0	no	0.767	0.132
Thick ungated	10.0	no	0.467	-0.0191
Thick ungated	25.0	no	0.017	-0.246
Thick ungated	50.0	no	-0.725	-0.6251
Thick ungated	10.0	yes (where archived)	0.467	-0.0191
ACWMI (IS-frozen)	0.0	no	0.654	0.0764
ACWMI (IS-frozen)	10.0	no	0.357	-0.0714
ACWMI (IS-frozen)	25.0	no	-0.086	-0.2935
ACWMI (IS-frozen)	50.0	no	-0.818	-0.6644
ACWMI (IS-frozen)	10.0	yes (where archived)	0.357	-0.0714
Momentum always	0.0	no	0.101	-0.2019
Momentum always	10.0	no	-0.198	-0.3522
Momentum always	25.0	no	-0.646	-0.578
Momentum always	50.0	no	-1.386	-0.9552
Momentum always	10.0	yes (where archived)	-0.198	-0.3522
Always long	0.0	no	-1.399	-1.1568

policy	cost_bps	funding	Sharpe	CE
Always long	10.0	no	-1.402	-1.1586
Always long	25.0	no	-1.407	-1.1613
Always long	50.0	no	-1.415	-1.1659
Always long	10.0	yes (where archived)	-1.402	-1.1586

Table 8. Transaction-cost and funding sensitivity (OOS).

7.6 Selective prediction on a sparse archive

IS calibration selects minimal additional index gating (frozen ACWMI < 0.25 binds on $\sim 1\%$ of days beyond content stand-asides) and the production WMI < 0.2 rule abstains 100%: quality thresholds are not portable across information-set supports. The measured epistemic calibration penalty is high (ECP = 0.688 at conf > 0.7 , WMI < 0.2): the classifier is often confident while the world is thin — the state ECP is designed to flag. All actions bind to named calculator evidence (EAR = 1, UCR = 0).

Read Section 7 as *content validation for the world-model runtime*, not as a frictionless trading mandate: (i) compiled content is economically nonempty and LOBO is content-dominant; (ii) mechanism – momentum isolates band content; (iii) the long-span audit rules out hidden return-rule alpha; (iv) absolute CE is cost-fragile—trading numbers prove content, they do not proclaim a Holy Grail; (v) WMI/ACWMI and Compiled-versus-Raw remain first-class runtime objects for denser archives and live LLM tests.

8. AI-consumer protocol (first-class interface)

The runtime is built for agents that condition on $\mathbf{s}_t^{\text{AI}}$. The replication package pre-registers a within-model Compiled versus Raw protocol, $\Delta_m = V_m(\text{Compiled}) - V_m(\text{Raw})$, with frozen prompts, temperature 0, and action schema {bullish, bearish, neutral, abstain}. In this framing the protocol is a *first-class interface test*: does the compiled world change consumer decisions relative to a raw feed? Offline mocks keep the protocol executable without API keys; live vendors may be substituted later. We do *not* claim live LLM trading alpha. Section 7 complements the interface by showing the world bundle carries economically nonempty band content for a transparent rule-based consumer.

Offline mock suite on the current OOS window: mean $\Delta\text{CE}(\text{Compiled-Raw}) = 0.0552$; models = mock-compiled-aware, mock-momentum, mock-noisy. Interpret as protocol smoke / calibration check, not as unconditional LLM alpha.

9. Robustness, threats, and the systems agenda

Right-censored bands. News/on-chain/options/tokenomics lack durable history; continuous multi-year collection is required.

Content vintages. Band-content identification rests on a single 400-day archive window; extend content calendars to multiple regimes.

Natural outages. Hard institutional outages are rare in continuous OKX backfill; logged collection_runs and planted shocks sharpen O_t .

Snapshot density. Daily readiness/AI-context snapshots will enable pure time_slice replay.

External validity. Expand beyond ten liquid names; the 2017–2026 audit anchors the return core only.

Multiple testing. The headline contrast is a single pre-specified comparison; the menu-wide reality check ($p = 0.144$) is reported alongside; add stationary-bootstrap confirmation.

Timing / ACWMI disclosure. Features for earning r_t are evaluated at $(t-1)$ 23:59. Live API bundles disclose $\text{acwmi_input_source} \in \{\text{paper_engines}, \text{production_proxy}\}$; paper empirics use engine (S,C).

AI-consumer limits. Offline Compiled–Raw mocks validate the runtime interface; they are not live multi-vendor LLM alpha. Support-matched abstention calibration and live consumers are first-order systems agenda items.

10. Conclusion

We build a financial-market world-model runtime: it compiles raw, asynchronous evidence into an LLM-analyzable, abstention-aware, and tradeable state $\mathbf{s}_t^{\text{AI}} = \sigma(\Pi_t(\mathbf{s}_t^{\text{raw}}))$. Epistemic observations, reconstruction bounds, WMI/ACWMI abstention, LOBO content/gating, and quality-tagged bundles are the semantics of that runtime; the laboratory (PIT, O_t , paper snapshots, AI bundles, Compiled-versus-Raw) is the system that implements them.

Trading numbers in this paper are not a strategy Holy Grail. Vintage-safe macro and stablecoin-flow content raise OOS CE relative to same-input momentum ($\Delta\text{CE } 0.334$, $p = 0.034$); LOBO is content-dominant; a long-span audit without band archives finds no hidden return-rule alpha. Absolute CE is cost-fragile. Those facts show that the compiled world has content—the necessary property of a world model worth serving to AI consumers. The agenda is thicker measurement and stronger consumers: multi-year vintaged histories, logged O_t , denser snapshots, support-matched abstention, and live Compiled-versus-Raw LLM tests under the frozen schema.

Appendix A. Notation

Symbol	Meaning
$O_{j,t}$	Epistemic observation
B_t, U_t, H_t	Breadth, stability, honesty
$WMI_t / ACWMI_t$	Production / regime-conditional index
Pi_t	Compilation operator
ECP / MIG	Calibration penalty / band information gain
Φ_t ; EAR/UCR/EV	Explanation set and metrics
O_t	Availability shock (DAG)

Table A1. Core notation.

Appendix B. Evidence bands and laboratory map

The laboratory organizes collectors into evidence bands that feed readiness and AI context. Table B1 lists band-level inventory used by the paper’s readiness weights; Table B2 samples logic modules.

Band	Module	Required	Asset weight	Role
exchange	exchange_data	Yes	0.3	Microstructure, prices, book, funding, basis
macro	macro_data	Yes	0.05	Macro vintages / rates / liquidity proxies
news	news_data	Yes	0.15	News and narrative evidence
event_calendar	event_calendar_data	Yes	0.1	ETF / unlock / event calendar
onchain	onchain_data	Yes	0.15	On-chain activity / TVL proxies
tokenomics	tokenomics_data	Yes	0.1	Unlocks / supply-side tokenomics
options	options_data	Yes	0.1	Options surface / gamma walls
alternative	alternative_data	No	0.05	Alt data (stablecoins, GitHub, trends)
perpetual_dex	perpetual_dex_data	No		Perp DEX funding / OI
onchain_address	onchain_address_data	No		Address-level on-chain labels
dex_liquidity	dex_liquidity_data	No		DEX TVL / tick liquidity
gas_network	gas_network_data	No		Gas / network congestion
governance	governance_data	No		DAO governance events

Table B1. Evidence bands (excerpt).

Module (1)	Module (2)	Module (3)
ai_market_context	alpha_decay	anomaly_detection
asset_readiness	contagion_risk	cross_asset_analysis
cross_venue_arbitrage	defi_stress	depth_regime
event_probability	exchange_comparison	feature_standardization
flow_decomposition	funding_rate_model	holder_behavior_analysis
liquidation_cascade	liquidity_analysis	liquidity_regime
logic_pipeline	macro_context	market_breadth
market_sentiment_composite	market_structure	miner_pressure
narrative_regime	news_sentiment	onchain_lead_lag
pipeline_latency	portfolio_risk	

Table B2. Logic modules (excerpt, 3-column grid).

Theory object	EvoQuant implementation	Role in paper
Market state compilation	data_layer collectors + logic_pipeline DAG	Core system
Breadth	43 domains / 13 audit bands / asset_readiness	Hierarchical B_hier
Stability	pipeline_latency freshness summary	U in WMI/ACWMI
Honesty	quality_flag + exclusion/contamination	Continuous H_cont
Signal integrity	alpha_decay half-life/crowding/surprise	S factor
Consistency	cross-band directional agreement	C factor
Mechanism engines	regime/cascade/contagion/flow/vol	Psi_mech
Degraded operation	DegradationManager levels	Resilient honesty
Abstention	should_ai_abstain + AC policy	State-dependent refusal

Table B3. Theory-to-implementation map.

Appendix C. Additional empirics

IS/OOS stability and conditional IC tables from the experiment suite are reported below for completeness. They complement the main CE/Sharpe horse-race and should be read under the same frozen-threshold protocol.

split	policy	Sharpe	CE
IS	WMI	0.0	0.0
IS	ACWMI	-0.2603	-0.4467
IS	Mom	-0.2153	-0.4292
OOS	WMI	0.0	0.0
OOS	ACWMI	0.6535	0.0764
OOS	Mom	0.1009	-0.2019

Table C1. IS/OOS stability.

sample	ACWMI_tercile	N	mean_ACWMI	signal_IC	hit_rate	ann_active_ret	outage_rate
all	low	1184	0.518	0.00515	0.575	1.8812	0.247
all	mid	1183	0.597	0.00281	0.568	1.025	0.006
all	high	1183	0.706	-0.00142	0.495	-0.5182	0.0
no_outage	low	1084	0.545	0.00367	0.534	1.3408	0.0
no_outage	mid	1082	0.606	0.00227	0.571	0.8288	0.0
no_outage	high	1084	0.712	-0.00106	0.504	-0.3878	0.0

Table C2. Conditional IC.

date	ready	stale	missing	freshness	klines	readiness	ai_ctx
2025-07-01	1	0	10	partial	missing	missing	missing
2025-08-01	1	0	10	partial	missing	missing	missing
2025-09-01	1	0	10	partial	missing	missing	missing
2025-10-01	1	0	10	partial	missing	missing	missing
2025-11-01	1	0	10	partial	missing	missing	missing
2025-12-01	1	0	10	partial	missing	missing	missing
2026-01-01	1	0	10	partial	missing	missing	missing
2026-02-01	1	0	10	partial	missing	missing	missing
2026-03-01	1	0	10	partial	missing	missing	missing
2026-04-01	1	0	10	partial	missing	missing	missing
2026-05-01	1	0	10	partial	missing	missing	missing
2026-06-01	1	0	10	partial	missing	missing	missing

Table C3. time_slice monthly grid (summary; analytics snapshots remain sparse historically).

Appendix D. Reproducibility

make paper-ai-wm | make paper-full | make paper-lab | python3 pdf/sci/generate_full_manuscript_pdf.py
 --variant ai-wm. Artifacts: pdf/data/pit_multiband_panel.csv; pdf/sci/main_ai_wm.tex;
 pdf/sci/main_ai_world_model.pdf; sibling JF draft pdf/sci/main_jf_rfs.tex.

*Draft status: AI-for-finance / systems framing of the same runtime + PIT content-validation empirics. Next: live
 Compiled-versus-Raw LLM tests, support-matched abstention calibration, and thicker vintaged histories.*

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